

QUANTIZATION CHANGES THE DATA: FULL-CONTEXT POST-TRAINING QUANTIZATION FOR VISION-LANGUAGE-ACTION POLICIES

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Paper under double-blind review

ABSTRACT

Post-training quantization (PTQ) usually calibrates a network on fixed teacher data and evaluates local reconstruction error. This formulation is incomplete for closed-loop vision-language-action (VLA) policies: quantization perturbs actions, altered actions change future observations, and the compressed policy therefore changes the input and activation-range distributions on which its own numerical fidelity depends. We call this the *full-context gap*. DYPAC-VLA addresses three coupled forms of missing context: temporal control context through physical-sequence divergence, whole-policy context through an exact-budget precision audit that may abstain, and deployment context through per-forward channelwise A8 scaling. Local layer interventions only propose masks; complete-policy evidence under teacher- and candidate-induced observations decides whether to accept a proposal or retain the initializer. On 50 RoboCasa365 tasks and 2,500 GR00T N1.5 episodes per configuration, DYPAC-VLA achieves **54.0%** success at $2.22\times$ static compression, compared with 30.4% for size-matched QuantVLA W4A8 (Holm-adjusted $p < 10^{-4}$) and 55.1% for FP16; the current test does not establish equivalence to FP16. The audit abstains on GR00T and retains its initializer. In a paired 500-episode intervention, a calibrated frozen A8 table obtains 0/500 successes, whereas input-conditioned A8 obtains 269/500. On $\pi_{0.5}$, the compressed policy obtains 27.7% success at $2.702\times$ static compression. These results expose fixed-distribution calibration as a concrete failure mode and motivate policy-level, abstaining evaluation for VLA compression.

1 INTRODUCTION

Vision-language-action (VLA) models connect visual perception and language-conditioned reasoning to continuous robot control Brohan et al. (2023); Octo Model Team et al. (2024); Kim et al. (2024); Black et al. (2024); Physical Intelligence et al. (2025); NVIDIA et al. (2025). Their large multimodal backbones and iterative action generators make static storage and memory bandwidth important deployment constraints. Post-training quantization (PTQ) is attractive because it reduces numerical precision without retraining the policy Xiao et al. (2023); Lin et al. (2024a).

Most PTQ formulations treat calibration data as exogenous: collect a fixed teacher buffer, minimize tensor error on that buffer, and deploy the resulting network. Closed-loop control breaks this assumption. Quantization changes an action; that action changes the next state and observation; the new input changes activation ranges and subsequent quantization error. The object being compressed is therefore not merely a network evaluated on fixed data, but a policy operating on the state distribution induced by its own quantized actions. We call the missing dependence between numerical error and policy-induced data the *full-context gap*.

The full-context gap has three coupled axes. *Temporal control context* is lost when local MSE ignores coherent prefix accumulation, nonlinear SE(3) composition, replan stitching, gripper timing, and rare catastrophic sequences. *Whole-policy precision context* is lost when a layer score decides precision without testing the complete mixed-precision network in which that layer operates. *Deployment-distribution context* is lost when a frozen activation table is assumed to cover observations, flow steps,

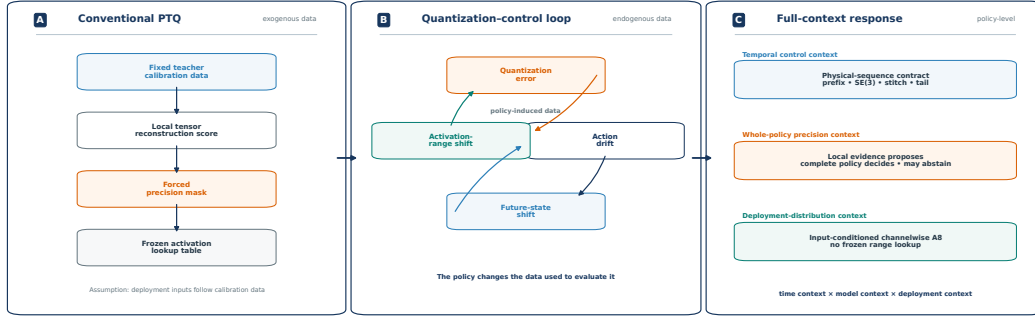


Figure 1: The full-context gap in VLA quantization. Conventional PTQ treats calibration data as fixed, but a quantized controller changes actions, future states, and the activation ranges on which later error depends. DYPAC-VLA restores temporal context with a physical-sequence contract, whole-policy context with an abstaining acceptance audit, and deployment context with input-conditioned activation ranges. Dataset-specific audit outcomes are reported in Section 4.3.

and states induced by the compressed policy. These are not three unrelated failure cases: they are views of the same quantization–control feedback loop.

We propose DYPAC-VLA, a full-context PTQ framework that audits policies rather than isolated tensors. D_{PAC} supplies a sequence-level behavioral contract in physical control space. Local counterfactuals then propose exact-budget masks, but an abstaining full-policy audit accepts a mask only if complete-policy evidence passes task-level uncertainty and component-regression guards under both teacher- and candidate-induced observations. Finally, DYRANGE-A8 removes the frozen activation lookup by conditioning channelwise A8 scales on the current deployment input. The operative principle is simple: local evidence proposes, the complete policy decides, and insufficient evidence preserves the initializer.

Figure 1 contrasts the fixed-data PTQ assumption with the quantization–control feedback loop and the three contexts restored by DYPAC-VLA.

We evaluate on RoboCasa365 Nasiriany et al. (2026) using 50 seeds on each of 50 tasks. On GR00T N1.5, DYPAC-VLA obtains 54.0% task-macro success at 962,068,480 static bytes ($2.22 \times$ compression), compared with 30.4% for QuantVLA W4A8 and 55.1% for FP16. The paired improvement over QuantVLA is significant after Holm correction ($p < 10^{-4}$); the difference from FP16 is not significant ($p = 0.2929$), which does not establish equivalence. The GR00T audit abstains and retains M_0 . A paired intervention isolates the demonstrated deployment-context mechanism: the registered frozen A8 table obtains 0/500 successes, while input-conditioned A8 obtains 269/500 under the same weights. Under a matched four-flow-step protocol, the $\pi_{0.5}$ policy obtains 27.7% success at $2.702 \times$ compression; a separate 4,000-episode LIBERO matrix provides cross-benchmark evidence for both models. The current evidence does not establish that D_{PAC} is a superior rollout predictor, that every static quantizer fails, or that dynamic A8 improves latency or live memory.

Our contributions are threefold.

- (1) **Closed-loop PTQ formulation.** We identify the full-context gap: quantization changes actions, future observations, and hence the deployment distribution on which numerical fidelity is evaluated.
- (2) **Abstaining full-policy audit.** We formulate mixed-precision selection as an exact-budget acceptance problem. Local interventions generate candidates; physical control sequences under teacher- and candidate-induced contexts determine whether a candidate is accepted or the procedure abstains.
- (3) **Input-conditioned deployment and large-scale evidence.** We remove frozen A8 lookup tables through per-forward channelwise scaling and report 54.0% success at $2.22 \times$ static compression on 2,500 GR00T episodes, plus a complete 4,000-episode LIBERO transfer matrix.

2 RELATED WORK

Efficient VLA inference. Generalist VLA policies combine large multimodal encoders with continuous-control heads [Brohan et al. \(2023\)](#); [Octo Model Team et al. \(2024\)](#); [Kim et al. \(2024\)](#); [Black et al. \(2024\)](#); [Physical Intelligence et al. \(2025\)](#); [NVIDIA et al. \(2025\)](#). Efficiency methods use compact architectures, token reduction, temporal reuse, or architecture–quantizer co-design [Shukor et al. \(2025\)](#); [Zhang et al. \(2025\)](#); [Yang et al. \(2025\)](#); [Xu et al. \(2025\)](#); [Fang et al. \(2025\)](#). These approaches may reduce computation as well as storage, but alter the network or its execution. DYPAC-VLA instead preserves the pretrained checkpoint structure and studies static mixed-precision compression; we do not claim latency improvement for the eager fake-quant implementation.

VLA quantization. QuantVLA adapts distribution transformations to language and diffusion-action stacks [Zhang et al. \(2026\)](#); ActQuant allocates tensor precision from action contribution and action-aware curvature [Akbari et al. \(2026\)](#); and Ω -QVLA combines rotations with denoising-step scaling [Wang et al. \(2026b\)](#). DA-PTQ is closest to our control-level motivation because it uses trajectory motion drift for mixed-precision VLA quantization [Xu et al. \(2026a\)](#). These methods improve how a VLA is scored or transformed on calibration data. Our focus is complementary: the compressed policy changes the future observations on which its own fidelity is evaluated. D_{PAC} specifies physical-sequence deviation within this formulation, but the current experiments do not establish that it predicts closed-loop success better than DA-PTQ or simpler metrics.

VLM PTQ and mixed precision. Vision–language PTQ also exposes modality and cross-layer effects. VLM-PTQ uses residual-aware compensation and modality-aware channel treatment [Deng & Huang \(2026\)](#); VLMQ weights reconstruction by token saliency [Xue et al. \(2026\)](#); Q-VLM models cross-layer discretization dependencies [Wang et al. \(2024\)](#); and Bi-VLM retains salient weights at higher precision in an ultra-low-bit design [Wang et al. \(2026a\)](#). These methods target vision–language understanding rather than continuous control, but they reinforce that independent layer scores can miss interactions. Our additive counterfactual program therefore generates candidates only; complete-network evaluation audits the finite proposals that it receives.

Transformer and diffusion PTQ. GPTQ and AWQ use second-order information or activation salience for weight-only quantization [Frantar et al. \(2023\)](#); [Lin et al. \(2024b\)](#). Weight–activation PTQ reshapes difficult distributions through smoothing, calibration, and rotations [Xiao et al. \(2023\)](#); [Shao et al. \(2024\)](#); [Ashkboos et al. \(2024\)](#); [Lin et al. \(2024a\)](#); [Liu et al. \(2025\)](#); [Sun et al. \(2025\)](#). Diffusion PTQ further addresses timestep-dependent ranges and iterative error accumulation [Shang et al. \(2023\)](#); [Li et al. \(2023\)](#); [Wu et al. \(2024\)](#); [Chen et al. \(2025\)](#); [Zhao et al. \(2024\)](#). These approaches generally calibrate on a fixed sample distribution, even when ranges vary across diffusion steps. DYPAC-VLA builds on a Hessian-aware group-64 W4 substrate but asks a different question: how should PTQ be evaluated when quantization changes the policy-induced deployment distribution? Its deployment response removes the frozen A8 lookup; we do not claim a new weight-rounding primitive.

Mixed precision and behavioral sensitivity. Prior mixed-precision methods rank layers using hardware feedback, Hessian sensitivity, reconstruction loss, or diffusion-specific proxies [Wang et al. \(2019\)](#); [Dong et al. \(2019; 2020\)](#); [Hubara et al. \(2021\)](#); [Li et al. \(2021\)](#); [Zhao et al. \(2024\)](#). Representation similarities such as CKA and distribution divergences are useful diagnostics [Kornblith et al. \(2019\)](#); [Yin et al. \(2026\)](#), but a local feature discrepancy does not identify how errors accumulate in executed control or whether interacting precision choices remain acceptable in the complete policy. We therefore separate proposal from acceptance: local evidence generates candidates, while an abstaining full-policy audit retains final authority under the deployment byte budget.

3 METHOD

Overview. DYPAC-VLA converts a frozen FP16 VLA into one static W4/FP16 policy while accounting for the data distribution induced by that policy. Figure 1 organizes the method around three contexts that conventional fixed-data PTQ omits. Temporal context defines behavioral fidelity over composed physical action sequences; whole-policy context gives final authority to an abstaining, exact-budget policy audit rather than a layer ranking; and deployment context conditions A8 ranges

on the current input. The output is one frozen mask and one activation rule, without task routing or output correction.

3.1 PROBLEM SETTING AND DESIGN CONSTRAINTS

Let $f_\theta(o, \epsilon)$ denote a frozen FP16 VLA that maps observation o and generative noise ϵ to an action chunk $A \in \mathbb{R}^{H \times d}$. The target Linear layers are indexed by $\mathcal{L}$. A mask $M \in \{4, 16\}^{|\mathcal{L}|}$ chooses signed-nibble W4 or native FP16 for every target, and $Q_M(f_\theta)$ denotes the instantiated mixed-precision policy. Let $\mathcal{D}_M$ be the closed-loop distribution over observation and replan sequences induced when $Q_M(f_\theta)$ executes its own actions. The conceptual full-space target is

$$M^* = \arg \min_{M \in \{4, 16\}^{|\mathcal{L}|}} \mathbb{E}_{\tau \sim \mathcal{D}_M} [D_{\text{ctrl}}(Q_M(f_\theta), f_\theta; \tau)] \quad \text{s.t.} \quad C(M) \leq C_{\max}, \quad (1)$$

where D_{ctrl} compares the quantized and teacher policies in physical control space on sequence τ , and $C(M)$ is exact packed static-component storage. Unlike fixed-buffer PTQ, Equation 1 makes the evaluation distribution depend on the precision plan. Exhaustively collecting $\mathcal{D}_M$ for every mask is infeasible. We therefore use teacher-induced contexts $\mathcal{D}_T$ for low-cost screening, then collect candidate-induced contexts $\mathcal{D}_M$ only for shortlisted complete policies. Section 3.3 defines the finite class $\mathcal{C}(M_0)$ that is actually audited and the abstaining output $\widehat{M}$.

This distribution dependence follows from ordinary closed-loop error propagation. For a local linearization with action error $e_t = a_t^Q - a_t^T$,

$$\delta x_{t+1} = A_t \delta x_t + B_t e_t, \quad \delta x_T = \sum_{t=0}^{T-1} \Phi_{T,t+1} B_t e_t, \quad (2)$$

where $\Phi_{T,t+1}$ composes the intervening state Jacobians. Per-step MSE measures $\|e_t\|^2$ but ignores propagation, coherent error direction, and the observations reached after the deviation. This motivates a sequence-level control contract and candidate-state audit; it is not, by itself, evidence that a particular surrogate predicts task success best.

The full-policy audit is label-free: success and reward never enter D_{ctrl} . Task identifiers only stratify uncertainty estimates, selection noise is paired, and a second noise stream is reserved for post-freeze audit. The complete initializer pipeline is not label-free because a four-task success sweep chooses the upstream CKA/CS ratio. The output is one mask and one activation rule shared by every task and observation. No gradient update, task-conditioned routing, output correction, or FP16 bypass is permitted.

These constraints separate compression design from test-time adaptation. A candidate may exploit the FP16 teacher and unlabeled on-policy observations before deployment, but it cannot observe task success or change its precision state online. This makes a selected mask auditable as a fixed model artifact and prevents a favorable task mixture from being encoded as a runtime selector.

3.2 TEMPORAL CONTEXT: A PHYSICAL-SEQUENCE CONTRACT

D_{PAC} defines what behavioral preservation means for a chunked, replanning VLA policy. It moves the fidelity criterion from tensor reconstruction to temporally composed physical control; the current evidence uses it as a specification and acceptance statistic rather than claiming proven predictive superiority over MSE or prior surrogates.

Paired physical actions. For task-seed sequence u , let $A_{u,r,h}^P$ be the final action from policy $P \in \{T, Q\}$ at retained replan r and chunk index h . Both policies receive the same observation and initial action noise. We sort unique replan indices, inverse-normalize the final action chunks, and retain the complete $H = 16$ actions executed before the next replan. Thus a sequence with R_u retained replans contains $N_u = 16R_u$ ordered actions in $d = 12$ deployed coordinates, without overlapping forecasts or deduplication. Only complete, finite chunks recorded before rollout termination enter the buffer. For coordinate k , a single global FP16-only scale is

$$\tilde{s}_k = \max(1.4826 \text{ MAD}_k, 0.1 \text{ RMS}_k), \quad s_k = \max(\tilde{s}_k, 0.05 \text{ median}_j \tilde{s}_j, 10^{-4}). \quad (3)$$

It is estimated once over the entire selection buffer and is never re-estimated per task or sequence. This prevents a severely corrupted candidate from making its own error appear small by inflating the denominator.

Local and accumulated error. Flatten sequence u in control-time order and define $z_{u,t} = (a_{u,t}^Q - a_{u,t}^T)/s$. We use the exactly-zero Charbonnier penalty $\rho(z) = 2(\sqrt{1+z^2} - 1)$. The sequence-level local and prefix terms are

$$d_{\text{local}}^{(u)} = \frac{1}{N_u d} \sum_{t,k} \rho(z_{u,t,k}), \quad d_{\text{prefix}}^{(u)} = \frac{1}{N_u d} \sum_{t,k} \rho\left(\frac{\sum_{j=1}^t z_{u,j,k}}{\sqrt{t}}\right). \quad (4)$$

The $\sqrt{t}$ normalization preserves sensitivity to coherent bias. Alternating errors can cancel, whereas a small error with constant sign remains visible as its prefix grows. This distinction is important for action chunks because consistent translation or rotation bias can move the end effector away from the teacher trajectory even when per-step MSE is small.

Control-specific components. Three terms capture deviations that coordinatewise residuals miss. The pose term right-composes translation and rotation-vector increments in SE(3) and measures normalized teacher–candidate trajectory drift. The stitch term compares changes in the boundary jump between consecutive replans, while the gripper term compares both softened command state and event timing. Appendix A.2 gives their exact formulas, action semantics, edge cases, and evaluation order.

The five normalized components receive equal weight:

$$d_{\text{PAC}}^{(u)} = \frac{1}{5} (d_{\text{local}} + d_{\text{prefix}} + d_{\text{pose}} + d_{\text{stitch}} + d_{\text{grip}}), \quad (5)$$

$$D_{\text{PAC}} = \text{mean}_u d_{\text{PAC}}^{(u)} + \text{CVaR}_{0.9,u} (d_{\text{PAC}}^{(u)}).$$

Here u is a task–seed sequence of ordered replans. The outer CVaR gives the worst ten percent of sequences the same total weight as the mean, exposing rare catastrophic deviations without duplicating an observation-level tail term.

We retain D_{func} as a complementary local final/kinematic diagnostic on the same physical chunks; it is not a second closed-loop loss. Its exact specification is deferred to Appendix A.2.

3.3 WHOLE-POLICY CONTEXT: AN ABSTAINING PRECISION AUDIT

Mixed-precision selection is treated as a statistical acceptance problem, not merely a ranking problem. Layerwise evidence can propose a precision plan, but only the instantiated complete policy can accept it. When no candidate clears the declared evidence threshold, the audit abstains by retaining its initializer M_0 .

Calibration-derived initialization. The audit starts from a frozen seed mask M_0 . A predeclared CKA/CS initializer scores layerwise W4 interventions, applies action-impact and RMS/saturation guards, and emits byte-feasible complete masks. A four-task development sweep selects the fixed proxy ratio; its 100-W4/16-FP16 mask is frozen before the full-context search and formal evaluation. We audit this initializer once, without post-hoc restarts.

Context buffer. We collect FP16 on-policy observations from 12 tasks, with four each from Atomic-Seen, Composite-Seen, and Composite-Unseen. Three seeds per task and four replans per sequence yield 36 sequences and 144 observations. A hash of task and seed assigns early, middle, or late rollout windows, preventing the selector from seeing only easy initial states. Task identifiers stratify statistics, but the result remains one global mask.

Coordinate counterfactuals. Starting from M_0 , round one flips every W4 target to FP16 individually. Round two flips every target state around the round-one winner. Each intervention is instantiated as a real mixed-precision network and compared with its complete-mask background. The resulting conservative benefits feed an exact sparse Pareto 0–1 dynamic program under

$$C(M) = C_{\text{fixed}} + \sum_{i \in \mathcal{L}} C_i(M_i) \leq 1.10 C_{\text{QuantVLA}}. \quad (6)$$

Layer count is not a constraint because two layers can have different tensor and metadata sizes. The dynamic program is only a candidate generator. Additive counterfactual benefits cannot represent interactions between simultaneous layer changes.

Let $\mathcal{C}_{\text{DP}}(M_0)$ denote the masks emitted by this additive program and $\mathcal{C}_{\text{struct}}(M_0)$ the five predeclared structured proposals. The implemented candidate class is

$$\mathcal{C}(M_0) = \{M_0\} \cup \mathcal{C}_{\text{DP}}(M_0) \cup \mathcal{C}_{\text{struct}}(M_0), \quad (7)$$

not the exponential domain in Equation 1. Complete-policy evaluation can detect interactions only for masks that enter $\mathcal{C}(M_0)$.

Task-level minimax adjudication. For metric $q \in \{D_{\text{func}}, D_{\text{PAC}}\}$ and task cluster $g \in \{\text{all}, \text{Atomic}, \text{C-Seen}, \text{C-Unseen}\}$, let $\Delta_{q,t}(M)$ be the paired candidate-minus- M_0 task scalar. We define

$$U_{q,g}(M) = \frac{\text{mean}_{t \in g} \Delta_{q,t}(M) + \text{SE}_{t \in g}^{\text{jack}}[\Delta_{q,t}(M)]}{\text{mean}_{t \in g} q_t(M_{\text{base}})}, \quad J(M) = \max_{q,g} U_{q,g}(M). \quad (8)$$

The leave-one-task-out jackknife makes the task, rather than an observation, the uncertainty unit. A proposal is eligible only if $J(M) < 0$ and the unnormalized one-SE upper bounds for pose, stitch, and gripper are all non-positive. These are conservative control-deviation guards, not physical-safety certificates.

Candidate-state check and freeze. Teacher-state observations can underrepresent the states induced by a changed policy. We therefore regenerate four to eight replans for the three best eligible masks and define $J_{\text{final}} = \max(J_{\text{teacher}}, J_{\text{candidate}})$. With the same component guards, the frozen output is

$$\widehat{M} = \begin{cases} \arg \min_{M \in \mathcal{E}(M_0)} J_{\text{final}}(M), & \mathcal{E}(M_0) \neq \emptyset, \\ M_0, & \mathcal{E}(M_0) = \emptyset, \end{cases} \quad (9)$$

where $\mathcal{E}(M_0) \subseteq \mathcal{C}(M_0)$ contains byte-feasible proposals with $J_{\text{final}} < 0$ and non-positive pose, stitch, and gripper upper bounds. Ties favor smaller bytes, fewer FP16 layers, then lexicographic mask order.

3.4 DEPLOYMENT CONTEXT: INPUT-CONDITIONED ACTIVATION QUANTIZATION

Deployment context removes the assumption that a frozen calibration table covers the inputs later encountered by a closed-loop generator. The novelty claimed here is not a new rounding primitive: Hessian-aware W4 is the weight substrate, while the activation rule conditions its range on the current deployment input.

Weight quantization. For each W4 layer, 256 calibration observations form FP16 input second-order statistics $X^\top X$. A GPTQ-style sequential error-feedback pass selects clipping and quantizes signed weights in groups of 64 [Frantar et al. \(2023\)](#). Two signed four-bit codes are packed per byte. Per-group scales and packing metadata are retained, while the FP16 weight tensor is not. Row rotation is the identity, so the reported method does not depend on an unreported learned transform.

Input-conditioned activation range. Even a calibrated static activation table can become mismatched as observation, flow step, and policy-induced state change. Our static counterfactual uses independently calibrated 99.9th-percentile scales for each target layer, input channel, and applicable flow step. DYRANGE-A8 removes this lookup and recomputes a scale for every target Linear, forward call, and input channel:

$$s_c = \max\left(\frac{\max |x_{\dots,c}|}{127}, 10^{-6}\right), \quad q_c = \text{clip}(\text{round}(x_c/s_c), -127, 127). \quad (10)$$

The rule is deterministic and uses the input dtype. It has no calibration lookup, flow-step selector, task selector, or candidate branch. Equation 10 is applied uniformly to every input, and runtime correction is disabled.

Exact storage scope. We report the byte sum of fixed model components, packed W4 codes, per-group scales and metadata, and native FP16 tensors. Activations, CUDA workspaces, simulator state, and eager fake-quant caches are excluded. Under this common scope the final GR00T plan contains 100 W4 and 16 FP16 targets and occupies 962,068,480 bytes. The $\pi_{0.5}$ anchor contains 121 W4 and 59 FP16 targets and occupies 1,634,828,288 bytes. These values establish static compression only. They do not imply latency or peak-memory improvements.

Table 1: RoboCasa365 results and exact static storage for GR00T N1.5 and $\pi_{0.5}$. Completed target-split rows use 50 paired scenarios per task.

Configuration	Allocation	Atomic SR $\uparrow$	C-Seen SR $\uparrow$	C-Unseen SR $\uparrow$	All SR $\uparrow$	Size (GiB) $\downarrow$	Comp. $\uparrow$
GR00T N1.5 (formal target split)							
FP16	—	75.6	41.9	45.3	55.1	1.993	1.00 $\times$
QUANTVLA W4A8	116 W4	49.6	19.6	19.8	30.4	0.898	2.22 $\times$
Uniform W6	116 W6	68.4	42.9	41.8	51.7	1.109	1.80 $\times$
Ω -QVLA W4A4 [‡]	180 W4	60.1	25.9	27.5	38.7	0.599	3.33 $\times$
QVLA-code Wavg4/A16 [§]	Ch. 0/2/4/8/16	Pending					
ActQuant 4.0 BPW/A16 [§]	Tensor IQ2-Q4	Pending					
DyPAC-VLA (Ours)[†]	100 W4	74.7	43.9	40.9	54.0	0.896	2.22$\times$
$\pi_{0.5}$ Physical Intelligence et al. (2025) (formal target split)							
FP16	—	58.3	14.0	2.1	26.2	4.113	1.00 $\times$
QUANTVLA W4A8	180 W4	56.1	12.8	1.8	24.8	1.388	2.96 $\times$
Uniform W6	180 W6	56.6	10.9	2.1	24.5	1.902	2.16 $\times$
Ω -QVLA W4A4 [‡]	252 W4	49.6	10.4	1.0	21.5	1.307	3.27 $\times$
QVLA-code Wavg4/A16 [§]	Ch. 0/2/4/8/16	Pending					
ActQuant 4.0 BPW/A16 [§]	Tensor IQ2-Q4	Pending					
DyPAC-VLA (Ours)[†]	121 W4	59.6	15.4	4.3	27.7	1.523	2.70$\times$

C-Seen/C-Unseen denote Composite-Seen/Composite-Unseen. All completed $\pi_{0.5}$ RoboCasa365 rows use four flow steps. [†]For each Ours row, the four tasks used to select the initializer ratio are included in the 50-task aggregate. GR00T ours uses 962,068,480 bytes; $\pi_{0.5}$ ours uses 1,634,828,288 bytes. [‡] Ω -QVLA uses model- and task-set-specific calibration. [§]QVLA/ActQuant [Xu et al. \(2026b\)](#); [Akbari et al. \(2026\)](#) formal extensions are pending. Compression denotes static storage, not PyTorch/C++ runtime memory or latency.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmark and models. We evaluate GR00T N1.5 and $\pi_{0.5}$ [Physical Intelligence et al. \(2025\)](#) on RoboCasa365 v1.0.1 [Nasiriany et al. \(2026\)](#): 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen tasks. Each completed row uses 50 fresh scenarios per task and official horizons (2,500 episodes). All target-split rows use four flow steps and shared task-seed-replan noise; formal paired claims are restricted to registered significance families.

Configurations and storage. FP16 is the teacher; QUANTVLA W4A8 [Zhang et al. \(2026\)](#), Uniform W6, and Ω -QVLA W4A4 [Wang et al. \(2026b\)](#) are low-bit references. The GR00T DyPAC-VLA plan has 100 W4 and 16 native-FP16 layers without output correction. Exact static bytes include fixed components and packing metadata, but not activations, workspaces, simulator state, or fake-quant caches. Its 962,068,480 bytes are 0.998 $\times$ the QUANTVLA cell and 2.22 $\times$ smaller than FP16.

Statistics and evidence gate. We report task- and split-macro success and exact paired two-sided McNemar tests with Holm correction. The headline uses 50 seeds per task; component tests use seeds 0–9 on all 50 tasks. Frozen manifests bind plans, code, environments, coverage, and hashes; any failed gate suppresses the cell.

4.2 MAIN RESULTS

GR00T headline. DyPAC-VLA obtains **54.0%** success versus 30.4% for QUANTVLA, a 23.6-point paired gain (Holm $p < 10^{-4}$); it is higher/tied/lower on 48/1/1 tasks. FP16 obtains 55.1% ($p = 0.2929$ versus ours), which establishes neither equivalence nor superiority. Figure 2 shows the measured success-storage tradeoff and task-level consistency; it is not a total ordering.

$\pi_{0.5}$ transfer. Exact-budget pruning converts 41 protected $\pi_{0.5}$ layers to W4, giving 121/180 W4 layers, 1.523 GiB, and 2.702 $\times$ compression. The four-flow-step policy records 693/2,500 successes:

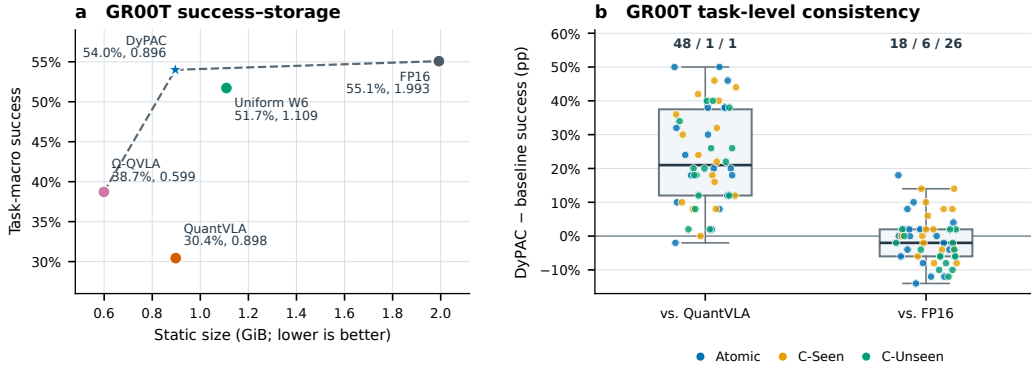


Figure 2: GR00T behavior-storage evidence on the 50-task RoboCasa365 target split. **(a)** Task-macro success versus exact static component storage (2,500 episodes per point); the dashed line connects non-dominated measured points and is not an interpolation. **(b)** Per-task differences for DyPAC-VLA on matched tasks and seeds; boxes show median and interquartile range, and counts are better/tied/worse tasks. The protocol-matched $\pi_{0.5}$ rows are reported separately in Table 1.

27.7% overall and 59.6/15.4/4.3% across the three groups. Its 1.6- and 2.9-point gaps over FP16 and QUANTVLA are descriptive; without a paired initializer comparison, this row isolates no audit gain.

Table 2: Cross-benchmark LIBERO transfer. Cells report four-suite mean success rate (%) and tightly packed static component size.

Configuration	GR00T N1.5		$\pi_{0.5}$	
	Avg. SR $\uparrow$	Size (GiB) $\downarrow$	Avg. SR $\uparrow$	Size (GiB) $\downarrow$
FP16	86.8	1.993	96.8	4.113
QUANTVLA W4A8	81.3	0.898	96.3	1.388
Uniform W6	70.5	1.109	96.8	1.902
Ω -QVLA W4A4 [†]	84.5	0.599	98.0	1.307
DyPAC-VLA (Ours)	93.8	0.948	97.3	1.788

The joint local gate covers 4,000 episodes across 40/40 exact-valid suite cells. [†]The displayed $\pi_{0.5}$ Ω -QVLA success value is experiment-owner attested. Ours size is the arithmetic mean of four suite-specific static plans. Appendix A.12 reports the full suite breakdown and protocol.

Cross-benchmark transfer. The separate 4,000-episode LIBERO matrix in Table 2 gives 93.8% at 0.948 GiB on GR00T and 97.3% at 1.788 GiB on $\pi_{0.5}$. Because Ω -QVLA reaches 98.0% at 1.307 GiB on $\pi_{0.5}$, this supports cross-benchmark and cross-model transfer, not universal dominance.

4.3 EVIDENCE FOR THE FULL-CONTEXT DIAGNOSIS

Table 3 separates offline admission from behavior. The paired objective rejects the registered per-layer/channel/flow-step static control, while DYRANGE-A8 remains close to the FP16-activation control. None of 116 flips or five structured masks passes the whole-policy rule, so the audit retains M_0 ; this shows neither selector gain nor global optimality.

Table 4 gives the matched 500-episode test: M_0 with DYRANGE-A8 obtains 53.8%, but the frozen table obtains 0.0% (Holm $p = 6.33 \times 10^{-81}$). This isolates failure of that static construction. Because the FP16-activation control is offline and compression excludes activations and workspaces, the result establishes neither A8 necessity nor runtime gains. Appendix A.7 gives supporting diagnostics.

The Local-MSE and unchecked-proposal masks each obtain 53.0% (−0.8 points; Holm $p = 1.0$), establishing neither equivalence nor a separate D_{PAC} /audit gain. Appendices A.6 and A.8 give mask and paired details.

Table 3: Core mechanism ablations on the frozen 12-task, 36-sequence context buffer. These are paired offline diagnostics, not closed-loop success rates; lower objective is better.

Factor	Controlled counterfactual	Audited evidence	Decision
Range	Static A8 vs. FP16-activation control	$J = 467.52$	Reject static A8
Range	Dynamic A8 vs. FP16-activation control	$J = 0.518$	Use DYRANGE-A8
Proposal audit	Every valid one-layer state flip	0/116 positive benefits	No eligible flip
Policy audit	5 structured mask alternatives	0/5 eligible; $J \in [3.97, 6.87]$	Abstain; retain M_0

J is the worst normalized one-standard-error upper bound over D_{PAC} , D_{func} , and task clusters. The FP16-activation control is diagnostic only. A mask is eligible only when $J < 0$ and pose, stitch, and gripper upper bounds are non-positive.

Table 4: Reduced closed-loop component ablation. Each configuration uses the same 50 tasks, seeds 0–9, paired noise, and official horizons. The reference is M_0 with DYRANGE-A8 on the same 500 episodes.

Configuration	W4/FP16	SR	Δ (pp)	Holm p
M_0 + DYRANGE-A8	100/16	53.8	reference	–
Frozen-range A8	100/16	0.0	–53.8	$< 10^{-4}$
Local-MSE + DYRANGE-A8	81/35	53.0	–0.8	1.000
No final check + DYRANGE-A8	115/1	53.0	–0.8	1.000

SR is reported in percent and Δ in percentage points. This reduced experiment estimates directional effects and does not establish equivalence when a difference is not significant.

Protocol separation and deployment scope. Only the CKA/CS ratio uses outcomes on four retained development tasks. The subsequent 12-task audit sees paired FP16 actions but no success labels, after which the mask and analysis family are frozen. Eager fake quantization retains floating-point execution, so we claim no latency, live-memory, or physical-safety gain.

5 CONCLUSION AND LIMITATIONS

VLA quantization is a closed-loop, distribution-dependent problem: quantized actions change future observations and hence the data used to judge the policy. DYPAC-VLA restores temporal context with a physical-sequence contract, whole-policy context with an abstaining exact-budget audit, and deployment context with input-conditioned activation ranges, using one static W4/FP16 mask without task routing or output correction.

On GR00T, this yields 54.0% RoboCasa365 success at $2.22\times$ static compression, versus 30.4% for size-matched QUANTVLA W4A8 and 55.1% for FP16; the latter comparison does not establish equivalence. In a paired 500-episode intervention, the registered frozen A8 table obtains 0.0% and input-conditioned A8 53.8%. The GR00T audit abstains, and alternative masks isolate no separate mean-success gain.

The protocol-matched, four-flow-step $\pi_{0.5}$ point is 27.7% at $2.702\times$ compression, but its gaps and mask change remain descriptive. Missing tests include teacher-forced feedback isolation, D_{PAC} predictive validation, a closed-loop FP16-activation arm, FP16 non-inferiority, and real latency, memory, and physical-safety measurements. They bound the present claim and define the next experiments.

AI USE STATEMENT

Generative AI tools assisted experiment planning, software implementation and debugging, result aggregation, drafting and editing, reference and table formatting, and architecture-figure preparation. The authors reviewed all generated material, tested AI-assisted code, reran evidence and document gates, verified numerical claims against frozen manifests, and checked references against

primary sources. No experimental result was synthesized by a generative model. The authors take responsibility for the final content and artifacts.

REPRODUCIBILITY STATEMENT

Sections 3 and 4 specify the metric, full-context search, quantization, activation, evaluation, and statistical protocols. The supplementary artifact provides content-addressed plans, manifests, task/seed coverage, table generation, and claim–evidence audits. The headline numerical cells are regenerated from frozen summaries rather than entered manually.

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A SUPPLEMENTARY EVIDENCE

A.1 CALIBRATION-DERIVED INITIAL MASK

The GR00T initializer is produced before the full-context search. For every predeclared CKA and CS ratio, action-impact weights prioritize layers whose intervention changes the physical action output, while RMS and saturation guards keep unstable layers in FP16. A greedy byte-constrained solver and fixed-size layer flips produce nine complete candidate masks. Paired adjudication on 16 calibration observations selects one mask for each ratio without using success labels. A four-task development sweep then selects the 16:1 ratio by closed-loop macro success. This gives the fixed proxy $16(1 - \widehat{\text{CKA}}_i) + \widehat{D}_{\text{CS},i}$ and the 100-W4/16-FP16 mask used as M_0 .

The full-context stage is run once from the frozen M_0 . It does not try several initial masks and retain whichever performs best. This design prevents post-hoc restarts from becoming another source of selection bias. The exact initializer artifact and its calibration buffer hash are recorded in the artifact registry. The four ratio-selection tasks are included in the 50-task aggregate. A descriptive recomputation over the other 46 tasks gives 51.8% for ours, 27.6% for QuantVLA W4A8, and 52.8% for FP16. This slice is not a separately preregistered comparison, but it shows that the main pattern is not created by the four development tasks.

A.2 EXACT METRIC SPECIFICATION AND EDGE CASES

Sequence construction and action semantics. Each metric unit is a task-seed sequence u with R_u unique replans. Replans are sorted by their recorded integer index, and each contributes the complete inverse-normalized (16, 12) chunk executed before the next replan, giving $N_u = 16R_u$. Later unexecuted forecasts are not retained, adjacent chunks do not overlap, and predictions are not deduplicated. A sequence contains only complete chunks collected before success, termination, truncation, or the official horizon. Duplicate replans, a shape other than (16, 12), or any non-finite value fail the metric gate rather than being padded or imputed. Coordinates 1–3 are end-effector translation, 4–6 are a rotation vector, 7 is gripper close, and 8–12 are the deployed base/control coordinates. Equation 3 normalizes every coordinate separately.

Pose map. For $\xi = [v; \omega]$, let $\Omega = [\omega]_\times$, $\theta = \|\omega\|_2$, and

$$\begin{aligned} \text{Exp}(\xi) &= \begin{bmatrix} I + A\Omega + B\Omega^2 & (I + B\Omega + C\Omega^2)v \\ 0 & 1 \end{bmatrix}, \\ A &= \frac{\sin \theta}{\theta}, \quad B = \frac{1 - \cos \theta}{\theta^2}, \quad C = \frac{\theta - \sin \theta}{\theta^3}. \end{aligned} \quad (11)$$

The implementation uses Taylor limits for $\theta < 10^{-5}$. Right multiplication $T_t = T_{t-1} \text{Exp}(\xi_t)$ interprets each increment in the current end-effector frame. The logarithm uses the principal rotation angle and inverse left Jacobian $V^{-1} = I - \Omega/2 + c\Omega^2$, where $c = 1/12$ near zero and $c = \theta^{-2} - (1 + \cos \theta)/(2\theta \sin \theta)$ otherwise. The three control-specific components are

$$e_t^{\text{pose}} = \frac{\text{Diag}(s_{1:6})^{-1} \text{Log}((T_t^T)^{-1} T_t^Q)}{\sqrt{t}}, \quad d_{\text{pose}}^{(u)} = \bar{\rho}_t(e_t^{\text{pose}}), \quad (12)$$

$$d_{\text{stitch}} = \bar{\rho} \left(\frac{(A_{r+1,1}^Q - A_{r,H}^Q) - (A_{r+1,1}^T - A_{r,H}^T)}{s} \right), \quad (13)$$

$$d_{\text{grip}}^{(u)} = \frac{1}{2} \left[\bar{\rho}_t(g_t^Q - g_t^T) + \bar{\rho}_{t>1}((g_t^Q - g_{t-1}^Q) - (g_t^T - g_{t-1}^T)) \right], \quad (14)$$

where $g_t^P = \sigma(4a_{t,7}^P/s_7)$. Translation and rotation are normalized coordinatewise before averaging.

Functional diagnostic. For physical final chunks indexed by context i , define

$$\begin{aligned} e_i &= \frac{\|A_i^Q - A_i^T\|_F^2}{\max(\|A_i^T\|_F^2, 10^{-8})}, \quad d_{\text{final}} = \frac{\|A^Q - A^T\|_F^2}{\max(\|A^T\|_F^2, 10^{-8})}, \\ D_{\text{func}} &= d_{\text{final}} + d_{\text{kin}} + d_{\text{grip}} + 2 \text{CVaR}_{0.9}(d_{\text{solver}}). \end{aligned} \quad (15)$$

d_{kin} is one half the sum of the analogous relative errors restricted to translation and rotation coordinates; $d_{\text{solver}} = \text{mean}_i e_i$, and $\text{CVaR}_{0.9}$ averages the e_i values at or above their empirical 90th percentile. The registered D_{func} gripper term is the fraction of adjacent solver states whose active gripper changes ($> 10^{-4}$ in both policies) disagree in sign. The shared selection path records only final physical chunks and no internal solver trajectory, so this legacy D_{func} gripper term is zero there; gripper deviation still enters D_{PAC} through Equation 14 and the explicit component guard.

Evaluation order. For each candidate, we (1) validate paired records, sort replans, and inverse-normalize final chunks; (2) compute the global 12-coordinate scale once from all FP16 selection chunks; (3) evaluate Equations 4, 12, 13, and 14 per task-seed sequence; (4) form the five-component mean and worst-10% sequence mean in Equation 5; and (5) compute task-level candidate-minus- M_0 differences, leave-one-task-out standard errors, J , and the pose/stitch/gripper upper bounds. A one-replan sequence has $d_{\text{stitch}} = 0$, while a one-action sequence has a zero gripper-difference term. Acceptance requires strict $J_{\text{final}} < 0$, non-positive component upper bounds, and the exact byte constraint; equality fails closed.

A.3 FROZEN GR00T DECISION RECORD

The GR00T search starts from the frozen M_0 . All 116 valid one-layer flips have non-positive benefit under the cross-context score. The additive dynamic program therefore emits no beneficial half- or full-budget combination. Five additional structured proposals are evaluated as complete networks; every alternative has a positive paired minimax objective or violates a pose, stitch, or gripper deviation guard. The fail-closed output is unchanged M_0 . These facts record rejection only within $\mathcal{C}(M_0)$ and do not establish local optimality outside that finite class.

The final plan uses signed-nibble group-64 weights, identity row rotation, per-forward dynamic A8, and no ATM, OHB, error-folding, calibration bypass, or runtime selector. Its SHA-256 is recorded in the artifact registry and checked before table generation.

A.4 MASK DIAGNOSTIC ROBUSTNESS

The registered 50-episode diagnostic compares M_0 with uniform full-W4 under an otherwise identical runtime. M_0 obtains 56% versus 46%, with ten paired wins, five losses, and 35 ties ($p = 0.302$). An earlier 50-episode exploratory wave reverses direction (50% versus 52%); pooling both waves gives 53% versus 49% ($p = 0.557$). The diagnostic is therefore disclosed only as a robustness check and does not support a causal claim or enter the headline evaluation.

A.5 PAIRED STATISTICAL DETAIL

Against QuantVLA W4A8, 1,647 of 2,500 episodes are tied, 721 favor ours, and 132 favor QuantVLA. Exact two-sided McNemar testing gives $p = 6.30 \times 10^{-99}$. Holm adjustment over the registered comparison family gives $p = 1.89 \times 10^{-98}$. Against FP16, the corresponding counts are 1,889 ties, 292 wins, and 319 losses, with $p = 0.2929$.

A.6 REDUCED CLOSED-LOOP COMPONENT ABLATION

The component study uses all 50 tasks and ten paired seeds per task. It is disjoint from mask construction and was aggregated only after all 1,500 new episodes passed the coverage audit. The matched M_0 reference obtains 73.9%, 46.9%, and 38.1% on Atomic-Seen, Composite-Seen, and Composite-Unseen, respectively. Its overall success rate is 53.8%.

Frozen-range A8 obtains 0.0% on every split. All 500 episodes complete normally, with no crash or numerical-failure row. Relative to M_0 , it has no paired wins, 269 losses, and 231 ties. The Holm-adjusted McNemar value is 6.33×10^{-81} . This result is specific to the frozen table constructed under our calibration protocol and is not a claim that every static activation quantizer must fail.

Local-MSE selection obtains 72.8%, 43.8%, and 40.0% across the three splits, for 53.0% overall. It has 55 paired wins, 59 losses, and 386 ties against M_0 . The no-final-check proposal obtains 70.6%, 38.8%, and 47.5%, also giving 53.0% overall. It has 54 wins, 58 losses, and 388 ties. Both Holm-adjusted values are 1.0. The split reversal for the no-final-check proposal illustrates why

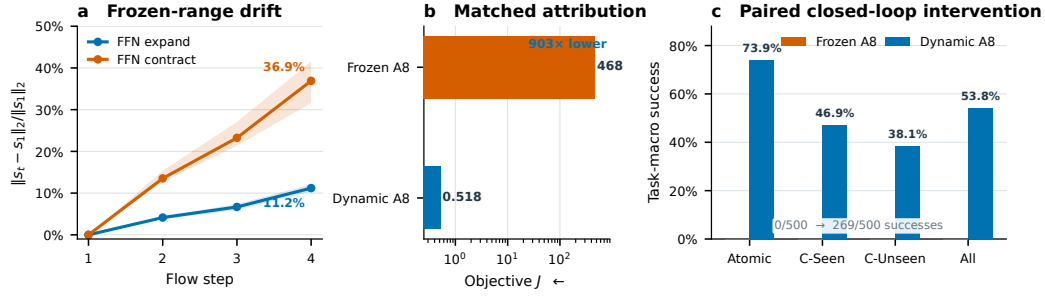


Figure 3: Evidence for per-forward A8 scaling. **(a)** Descriptive drift of calibrated per-channel ranges across flow steps. **(b)** Offline matched attribution relative to A16 under objective J (log scale). **(c)** Under the otherwise identical M_0 runtime, the registered static table obtains 0/500 successes and dynamic A8 obtains 269/500. This comparison does not cover every possible static activation quantizer.

the full 50-task result is preferable to conclusions drawn from only the two seen groups. Neither non-significant comparison is treated as evidence of equivalence.

A.7 DYNAMIC-RANGE EVIDENCE

Figure 3 separates mechanism diagnosis from behavioral evidence. Across the 31 applicable action-head layers, the median scale drift is 10.8%, 16.8%, and 24.9% at flow steps 2–4, and the maximum layer-step value is 49.0%. These frozen-table statistics do not by themselves establish a success effect. The matched offline attribution reduces J from 467.52 to 0.518, while the paired closed-loop intervention compares 0/500 successes for the registered static table with 269/500 for per-forward dynamic A8 under the otherwise identical M_0 runtime.

A.8 FULL-POLICY AUDIT OUTCOME

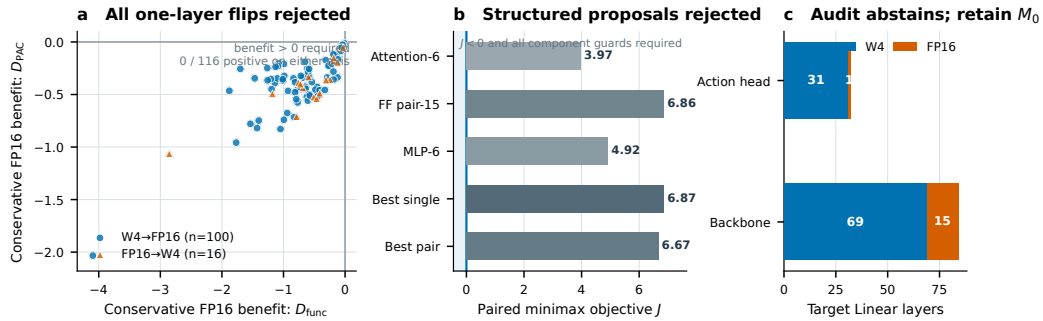


Figure 4: GR00T outcome of the abstaining full-policy audit. **(a)** Conservative FP16 benefits for all 116 valid one-layer counterfactuals; a positive value is required, but neither diagnostic is positive for any flip. **(b)** Each structured proposal has $J > 0$ and fails at least one component-deviation guard; acceptance requires $J < 0$ and all guards. **(c)** The audit therefore abstains and preserves the 100-W4/16-FP16 initializer. This finite-neighborhood decision is neither a separate measured success gain nor evidence of global optimality or physical safety.

Figure 4 makes the acceptance boundary explicit. Every valid one-layer precision flip has non-positive conservative benefit under both D_{func} and D_{PAC} . All five complete structured proposals also have positive minimax objective ($J = 3.97$ – 6.87) and fail the registered component-deviation gate. The full-policy audit consequently abstains and returns the unchanged 100-W4/16-FP16 M_0 . This visualization documents the decision only within the declared search neighborhood; no separate gain or global-optimality claim follows from it.

A.9 STORAGE ACCOUNTING

The GR00T FP16 static-component scope is 2,139,537,408 bytes. The final plan occupies 962,068,480 bytes, including 327,352,320 fixed bytes and packed mixed-precision target components, for $2.2239\times$ compression. This is $0.9982\times$ the QuantVLA cell of 963,772,416 bytes and satisfies the preregistered $1.10\times$ ceiling. The $\pi_{0.5}$ FP16 scope is 4,416,602,112 bytes. Its 1,634,828,288-byte plan gives $2.7016\times$ compression.

These values are static packed estimates. Activations, CUDA workspaces, simulator state, and eager fake-quant caches are outside the scope. Therefore no runtime speed or peak-memory conclusion follows from the compression ratios.

A.10 $\pi_{0.5}$ FORMAL RESULT

The $\pi_{0.5}$ transfer uses four flow steps on the complete RoboCasa365 target split, matching the completed reference rows, with 50 seeds on each of 50 tasks. The 121-W4/59-FP16 policy completes all 2,500 episodes without a missing, duplicate, or failed row. It records 536/900 Atomic-Seen, 123/800 Composite-Seen, and 34/800 Composite-Unseen successes, giving 693/2,500 overall. The corresponding task-macro rates are 59.6%, 15.4%, 4.3%, and 27.7% overall. The result supports a formal success-storage operating point at 1,634,828,288 bytes and $2.702\times$ compression. Cross-row differences remain descriptive because this experiment does not register a paired significance comparison with the other $\pi_{0.5}$ configurations.

A.11 EXTENDED ABLATION PROTOCOL

Table 5 registers the experiments needed to test claims that the completed evidence does not support. Every selection experiment will start from frozen checkpoints and manifests, preserve exact-byte accounting, report the resulting mask rather than force equal layer counts, and use paired task-seed evaluation. All entries remain pending and contribute no numerical evidence to the current paper.

Table 5: Registered pending experiments. These studies are not complete and provide no numerical evidence in the current paper.

Family	Variants	Controlled readout	Status
Full-policy audit behavior	Multiple initial masks and byte budgets; compare accepted masks with their initializers	Acceptance/abstention rate; paired success and byte deltas	Pending
D_{PAC} predictive validity	Rank a broad library of masks and activation variants using D_{PAC} , D_{func} , and MSE	Spearman/Kendall correlation with rollout success	Pending
D_{PAC} components	Remove prefix, pose, stitch, gripper, or CVaR one at a time	Rank stability, selected mask, paired success	Pending
Static A8 controls	Max, clipped percentile, per-flow, and sequence-aware static calibration	Paired success, objective, saturation rate	Pending
$\pi_{0.5}$ paired inference	Register the four-flow-step Ours-reference paired comparison family	Exact McNemar, Holm correction, and task-level deltas	Pending
QVLA/ActQuant extension	Complete the registered GR00T and $\pi_{0.5}$ formal rows	Coverage audit, task-macro success, exact bytes	Pending
Guard validation	Relate pose, stitch, and gripper deviation bounds to observed control failures	Calibration/AUROC; failure-category rates	Pending
Budget sensitivity	1.00, 1.05, and $1.10\times$ the QuantVLA static-byte cell	Independently selected success-storage points	Pending

Each row is a protocol commitment rather than a result. *Pending* means that no value from the experiment is used in the Abstract, main claims, or statistical comparisons. The guard-validation row evaluates surrogate control deviations and is not a physical-safety certification protocol.

A.12 PROTOCOL-REGISTERED LIBERO EXTENSION

Table 6: LIBERO comparison in success rate (%). Local size is tightly packed static model-component storage.

Configuration	Goal $\uparrow$	Spatial $\uparrow$	Object $\uparrow$	Long $\uparrow$	Avg. $\uparrow$	Size (GiB) $\downarrow$
GR00T N1.5						
FP16	88.0	87.0	96.0	76.0	86.8	1.993
QUANTVLA W4A8	89.0	85.0	79.0	72.0	81.3	0.898
Uniform W6	85.0	83.0	92.0	22.0	70.5	1.109
Ω -QVLA W4A4	87.0	86.0	89.0	76.0	84.5	0.599
DYPAC-VLA (Ours)	97.0	95.0	98.0	85.0	93.8	0.948
$\pi_{0.5}$						
FP16	95.0	100.0	98.0	94.0	96.8	4.113
QUANTVLA W4A8	97.0	97.0	98.0	93.0	96.3	1.388
Uniform W6	99.0	97.0	99.0	92.0	96.8	1.902
Ω -QVLA W4A4	100.0	99.0	97.0	96.0	98.0	1.307
QVLA (4.0 BPW) [†]	96.4	98.0	97.2	91.8	95.8	2.7
ActQuant (4.0 BPW) [†]	96.8	98.4	99.4	91.8	96.6	2.7
DYPAC-VLA (Ours)	99.0	100.0	96.0	94.0	97.3	1.788

The complete local matrix contains 4,000 episodes across 40/40 exact-valid cells. Every local suite value covers exactly ten tasks and held-out initial states 10–19. The corrected GR00T ours row uses five executed actions per replan and paired suite-keyed action noise. Local size includes packed weights and quantization metadata. For suite-specific masks, it is the four-suite mean. [†]The $\pi_{0.5}$ QVLA and ActQuant entries are the source-reported 4.0 Vision+LLM-BPW rows from ActQuant Akbari et al. (2026); their 2.7-GB values follow that source’s memory definition.

Table 6 extends rather than defines the central claim. The corrected GR00T DYPAC-VLA row obtains 97.0%, 95.0%, 98.0%, and 85.0% on Goal, Spatial, Object, and Long (93.8% average). Each local cell contains 100 episodes over ten tasks and held-out initial states 10–19; all 40 cells pass the joint gate, for 4,000 episodes with no missing or duplicate keys. The rerun executes five actions per replan with paired action noise. We report the corrected measurement without treating the change from the invalid run as an algorithmic gain.

On GR00T, QUANTVLA W4A8, Uniform W6, and Ω -QVLA W4A4 average 81.3%, 70.5%, and 84.5%, versus 86.8% for FP16. On $\pi_{0.5}$, the corresponding local low-bit rows average 96.3%, 96.8%, and 98.0%. Tightly packed static size is reported for every local row. The two 4.0-BPW $\pi_{0.5}$ anchors from ActQuant Akbari et al. (2026) are source-reported; its QVLA row reproduces QVLA Xu et al. (2026b) under the same 60-episode calibration budget. Neither enters local coverage, and their 2.7-GB cells are not compared with locally audited size.